



**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 1

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Subject Name : OBJECT ORIENTED ANALYSIS AND DESIGN

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Unit : I

Teacher's Sign :

Department Vision and Mission

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission:

- * To impart quality education in computer science and engineering by means of learning techniques and value-added courses.
- * To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.
- * To build leadership skills, ethical values and teamwork among the students.
- * To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

2 Marks

1. Differentiate between OOSA and OOSD

OOSA

- OOSA is primarily connected with understanding the problem domain and gathering the requirements.
- It's about analyzing the system from a high level perspective without worrying about the internal detail.
- Class diagrams and use-case diagrams are common artifacts.

OOSD

- OOSD is about translating the analysis model into a blueprint for building the system.
- It involves defining the structure, behaviors and interaction of objects in the system.
- Class designs, sequence diagram, state diagram, and other design, level artifacts.

2. Define prototype. Give the types of prototype.

- A prototype is object-oriented analysis and design is an early working model or sample of a software system or its components.
- It is built to visualize and test certain aspects of the system functionality, usability or design before the complete system is fully developed.
- Prototype help stakeholders such as developers and users.

Types:

- Throwaway prototype
- Evolutionary prototype
- Incremental prototype
- Extreme prototype.

3. What is the strength of Jacobson et al methodology?

- Use case driven
- Emphasis on object responsibilities.
- Clear separation between analysis and design
- Focus on iterative and Incremental development
- Incorporation of object-oriented concepts.
- Modeling at multiple levels.
- Unified Approach.
- Focus on communication and stakeholder involvement.
- Reuse of existing models.

4. What are the phases of OMT.

- object modeling (Analysis phase)
- Dynamic modeling (Behavioural phase)
- Functional modeling (Functional phase)

5. List the phases of SDLC.

The software development Life cycle (SDLC) typically follows a set of well-defined phases. These phases provide a structured approach for developing software using object-oriented principles.

- Requirements gathering Analysis
- object-oriented analysis
- object-oriented design
- Implementation (coding)
- Deployment
- Maintenance

6. Difference between verification and validation.

Verification

- It checks whether the system is being built correctly according to the specified design and requirement.
- It focuses on the internal consistency and correctness of the design, code and models.
- To ensure that the product the design specifications and that each part of the system is implemented as intended.

Validation

- It checks whether the system meets the user's needs and expectations.
- It focuses on the external behaviours and usability of the system.
- To ensure that the final product fulfills its intended purpose and requirements from the user's perspective.

7. Difference between static and dynamic models.

Static

- * It represents the structure of a system, showing the static relationships between its components.
- * To describe the system's data and organization without considering behaviour or interactions over time.
- * Focuses on the static aspects like classes, objects, attributes, methods and the relationships.

Dynamic

- * It represents the behaviour of the system, illustrating how objects interact and change state over time.
- * To describe the flow of control and data within the system as it operates.
- * Focuses on the dynamic behaviour, interactions, and state machine diagrams, activity.

8) What is UML?

- UML (Unified Modeling Language) is a standardized modeling language used in software engineering to visualize, specify, construct and document the structure and behaviours of software systems.
- It helps developers and designers to communicate complex system designs clearly and effectively.

9) Define Rational Unified Process and write four phases. RUP.

Rational Unified Process (RUP) is a software development framework that provides a structured and disciplined approach to designing, building, and deploying software systems.

It is iterative and incremental, allowing developers to refine and improve the system through multiple cycles.

Phases

- Inception phase
- Elaboration phase
- Construction phase
- Transition phase.

10) What is use case and write the symbol used in it.

A use case is a description of how users interact with a system to achieve a specific goal.

It defines the functional requirements of a system.

1. Actor: Represented by a stick figure.
2. Use case: Represented by an oval with the name of the use case inside.
3. System boundary: Shows ~~name~~ as a rectangle enclosing the use cases. The system name is usually written at the top.
4. Association: A solid line, connecting an actor to a use case, representing interaction.

11 Marks

1. Discuss object oriented system development Life cycle.

- The software development process consists of analysis design, implementation, testing and refinement phase which is used to transform the user needs to software solution that satisfies those needs.
- Software development can be viewed as a process the development itself is a process of change, refinement transformation or addition to the existing product.
- Within the process, it is possible to replace one sub process with a new one, as long as the new process has the same interface as the old one.
- With the method of change it is possible to adopt to a new process.
- The process can be divided into small interacting phases called sub processes.

→ Each sub process must have the following behaviour.

- * A description in terms of how it works.
- * Specification of the outputs to be produced.
- * Specification of the input required for process.

The software development process can be viewed as a series of transformations,

- * Transformation 1 (Analysis)
- * Transformation 2 (Design)
- * Transformation 3 (Implementation)

Transformation 1 (Analysis)

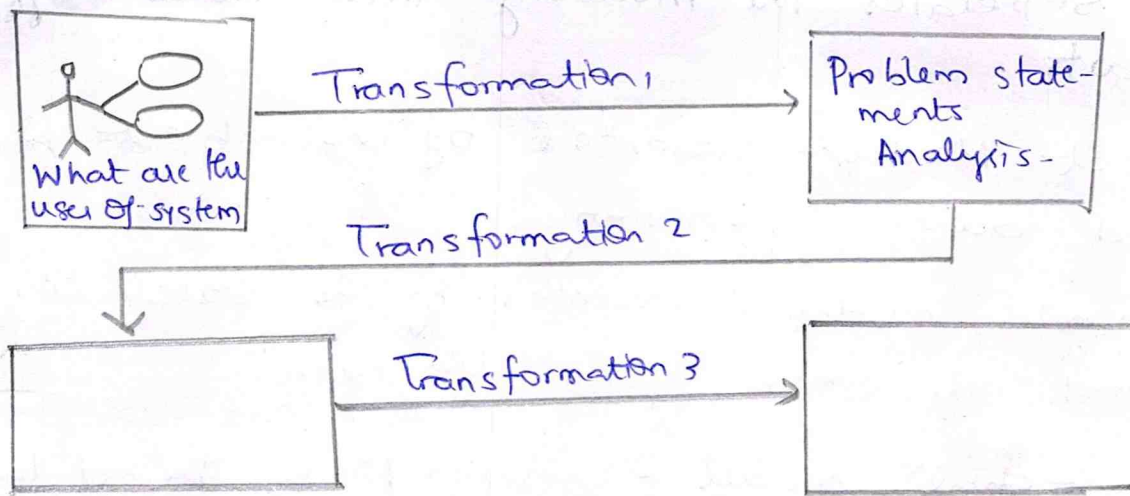
- * This translates the users need into system requirement and responsibilities.
- * This always gives an insight to the user's requirements.

Transformation 2 (Design)

- * This layer begins with a problem statement and ends with a detailed design that can be transformed into an operational system.
- * This transformation includes the build of software development activity, including the definition of how to build the software, its development and its testing.

Transformation 3 (Implementation)

- * This refines the detailed design into the system deployment that will satisfy the users needs.
- * It represents embedding the software product within its operation environment.



2) Describe Rumbaugh object modeling Technique with banking system.

- Rumbaugh et al method is well suited for describing the object model or the static structure of the system.
- The object modeling system (OMT) presented by Jim Rumbaugh and his workers describes as method for the analysis, design and implementation of a system using an object making up in system can be described using OMT dynamic model.
- A process description and consumer produce relationships can be expressed using OMT's functional model.

OMT consists of four phases.

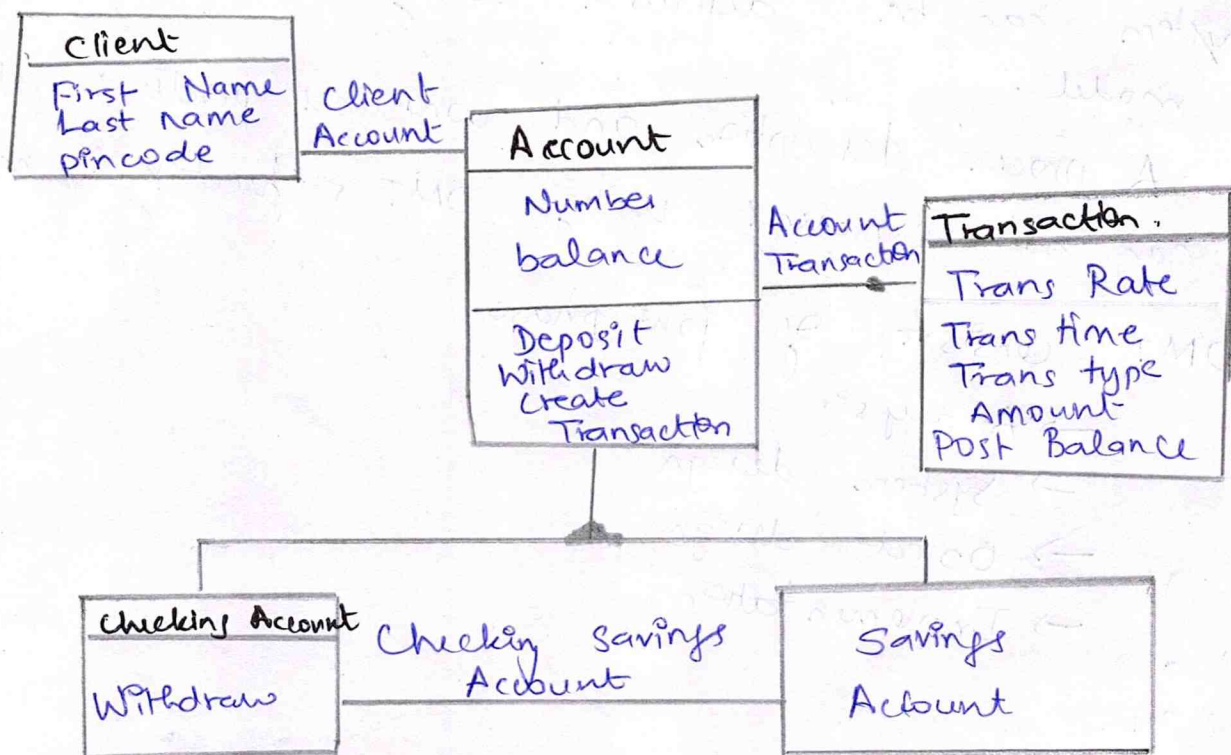
- Analysis
- System design
- Object design
- Implementation.

OMT separates the modeling into three different parts.

- * Object Model - presented by the object model and data dictionary.
- * Dynamic model - presented by the state diagram and the event flow diagram
- * Functional model - presented by the data flow and constraints.

object model

- * It describes the structures of objects in a system their identity, relationship to other objects attributes and operations.
- * The object diagram consists of class intercomm by association lines.
- * The association lines establish relationships among the classes.
- * Each class represents a set of individual objects.

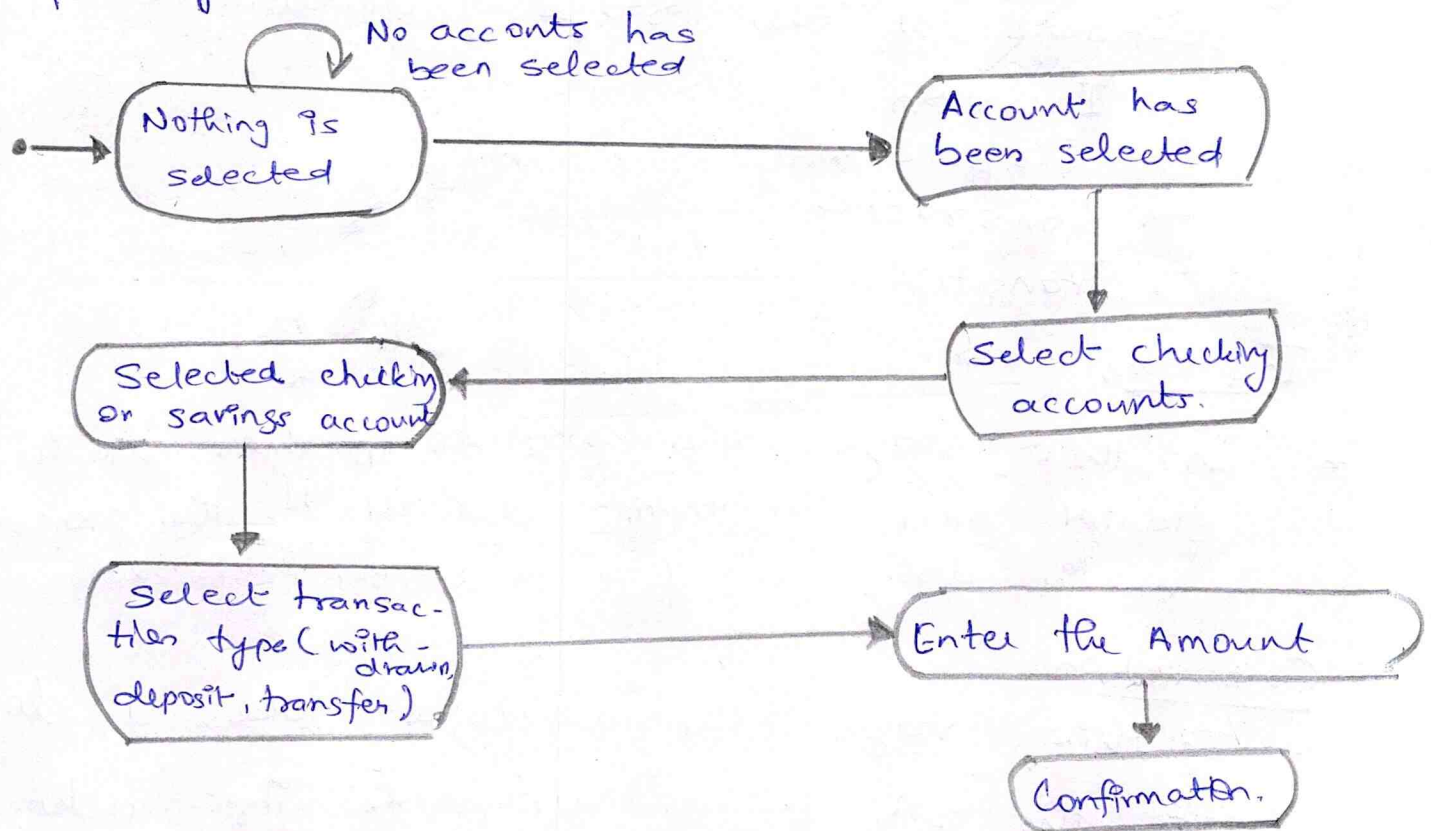


The OMT Dynamic Model

- * The OMT provides a detailed and comprehensive dynamic model. It is a network of states and events.
- * Each state receives one or more events, at which time it makes one or more transitions to the next state.
- * The next state depends on the current state as well as the events.

OMT Functional Model

- * The OMT data flow diagram (DFD) shows the flow of data between different processes in a business.
- * An OMT DFD provides business processes without method for describing business processes without focusing on the details of computer systems.



3. Explain detail about rational unified process with a neat diagram.

- * A life cycle provides a high level representation of the stages that a development project must go through to produce a successful system, but it does not attempt to dictate how the should be achieved, nor does it describe the activities that should be carried out at each stage.
- * A development method. I on the other hand, is much more prescriptive, often setting down in detail the task, responsibilities, processes, prerequisites and milestones for each stage of the project these of the chapter and in the bibliography.
- * Short for rational unified process, a based on UML. RUP organize for the development of software into four phases, each consisting of the one or more executable iterations of software at the stage of development.
 - Inception
 - Elaboration
 - Construction
 - Transition

Inception

- * In this stage, the project's business case is started and the team decides if the project is worth doing or if it is even possible.

Elaboration

- * In this stage, the developer take a close look the project to determine its architecture

Foundation and to evaluate the architecture in relation to the project.

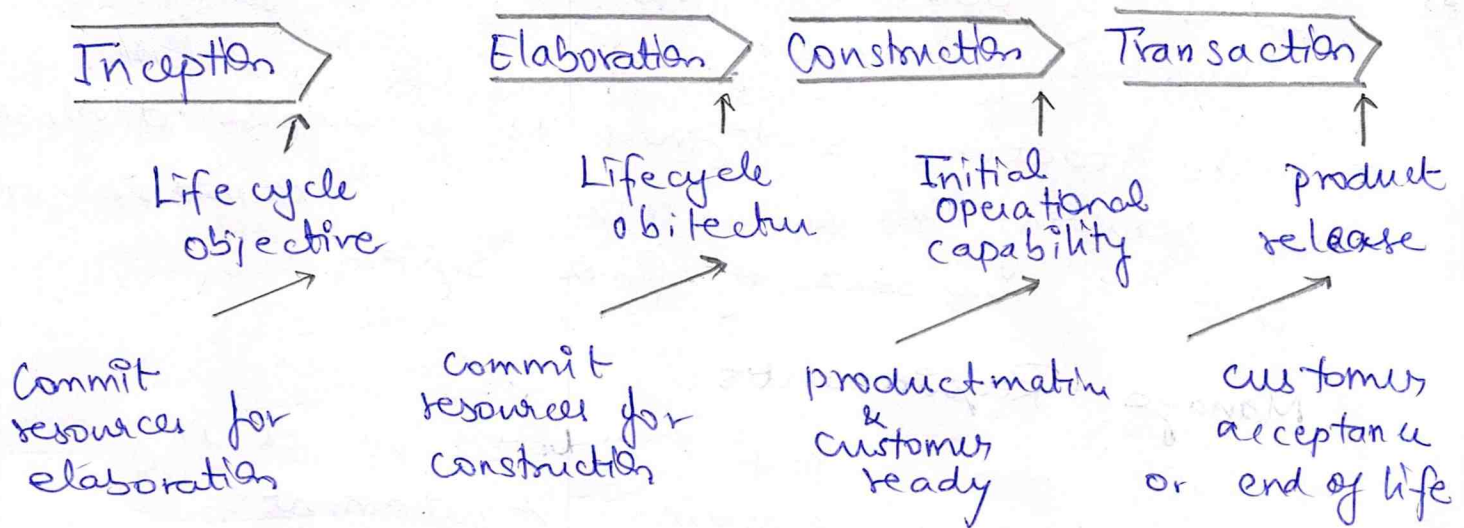
Construction

In this stage, the development of the project is completed. The application design is finished and all the source code is written.

Transaction

In this stage, fine tuning is performed, any final adjustments can be used on based feedback, usability or installation issues.

Phases of RUP



RUP phases

- * RUP is corporate not only modeling technique from the UML, but also guidelines, templates and tools to ensure effective and successful development of software systems.
- * The tools offered by RUP allow a large part of the development process to be automated.

* RUP is based on the following six best practices which have been formulated from experience on many industrial software projects.

- develop software iteratively
- manage requirements
- use component based architecture
- visually model software.
- verify software reality,
- control change to software.

Develop software iteratively

- * It is built on the central concept of iterative development and each of its phases define a series of additive tasks that may be performed on or a number of times.
- * Each iteration is defined as a complete development loop resulting in the release of an executable product that is a subset of the final system.

Manage Requirements:

- * RUP offers sound support for eliciting, organising and recording requirements.
- * Precise documentation of requirements facilitates consistency through the development process, which enhances the quality of the final system.

Use component based architecture

- * RUP provides the early identification and development of a system structure that is at the same time robust enough to ensure system reliability, and flexible enough to accommodate changes.

Verify software qualities

- * RUP provides the technique to support quality attributes of functionality, reliability and performance throughout the development process.
- * The RUP approach to quality is based on objective measure and criteria for success.

Visually model software

- * RUP is based around the unified modeling language (UML) a vehicle for development.
- * The technique offers by UML being with them all the advantage of visual modeling.

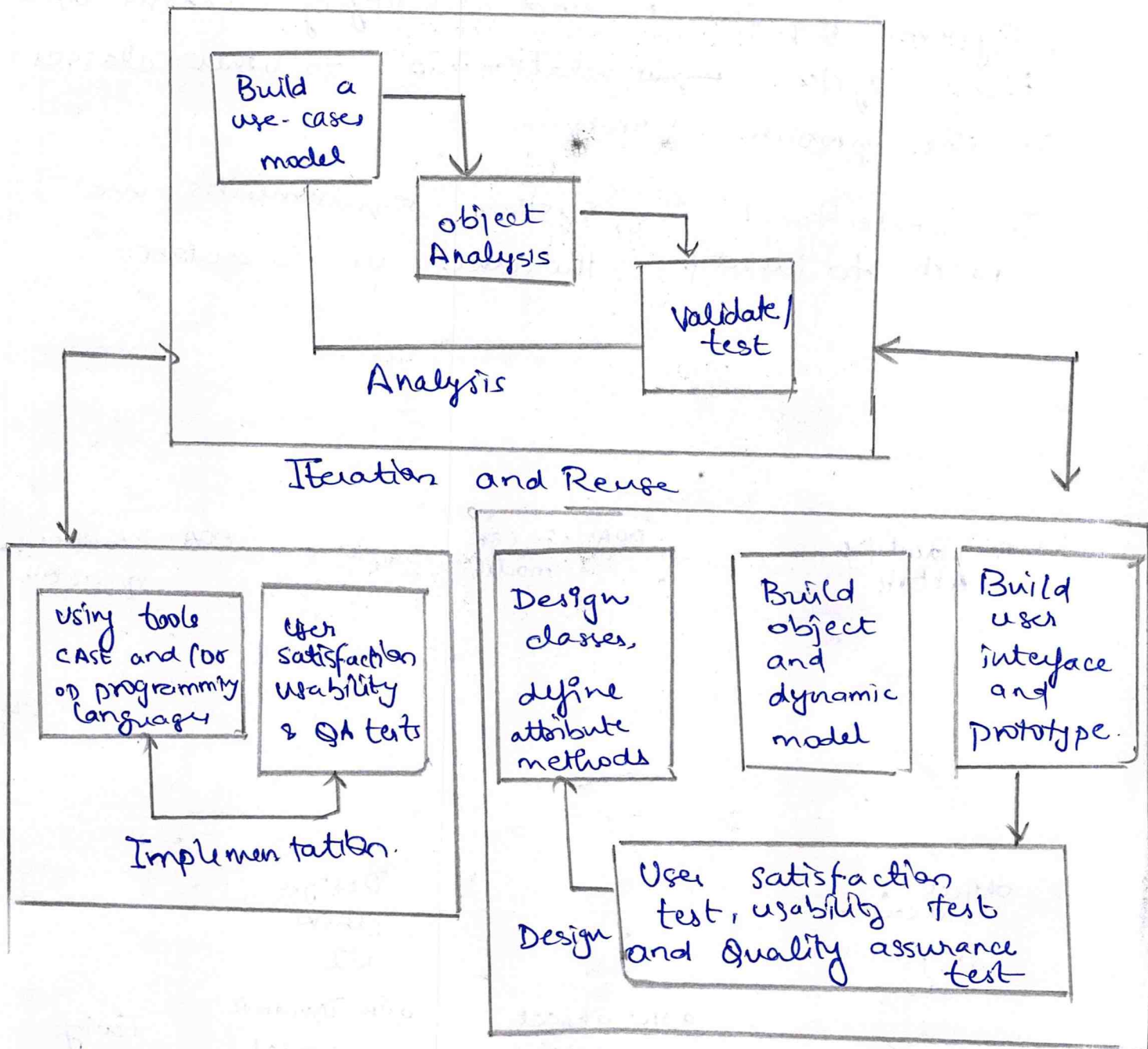
Control changes to software.

- * Change are the noun is a software developers projects to an effective development process must be able to monitor and control them.
- * RUP provide tools to do this and also supports the work of developers by offering protection is one area of development from changes that occur in another.

4) Discuss about object oriented system development Life cycle.

OBJECT ORIENTED SYSTEMS DEVELOPMENT

The object oriented software development Life cycle (SDLC) consists of three macro processes. Object oriented analysis, object - oriented design and object oriented implementation.



Object - oriented system development includes .

These activities .

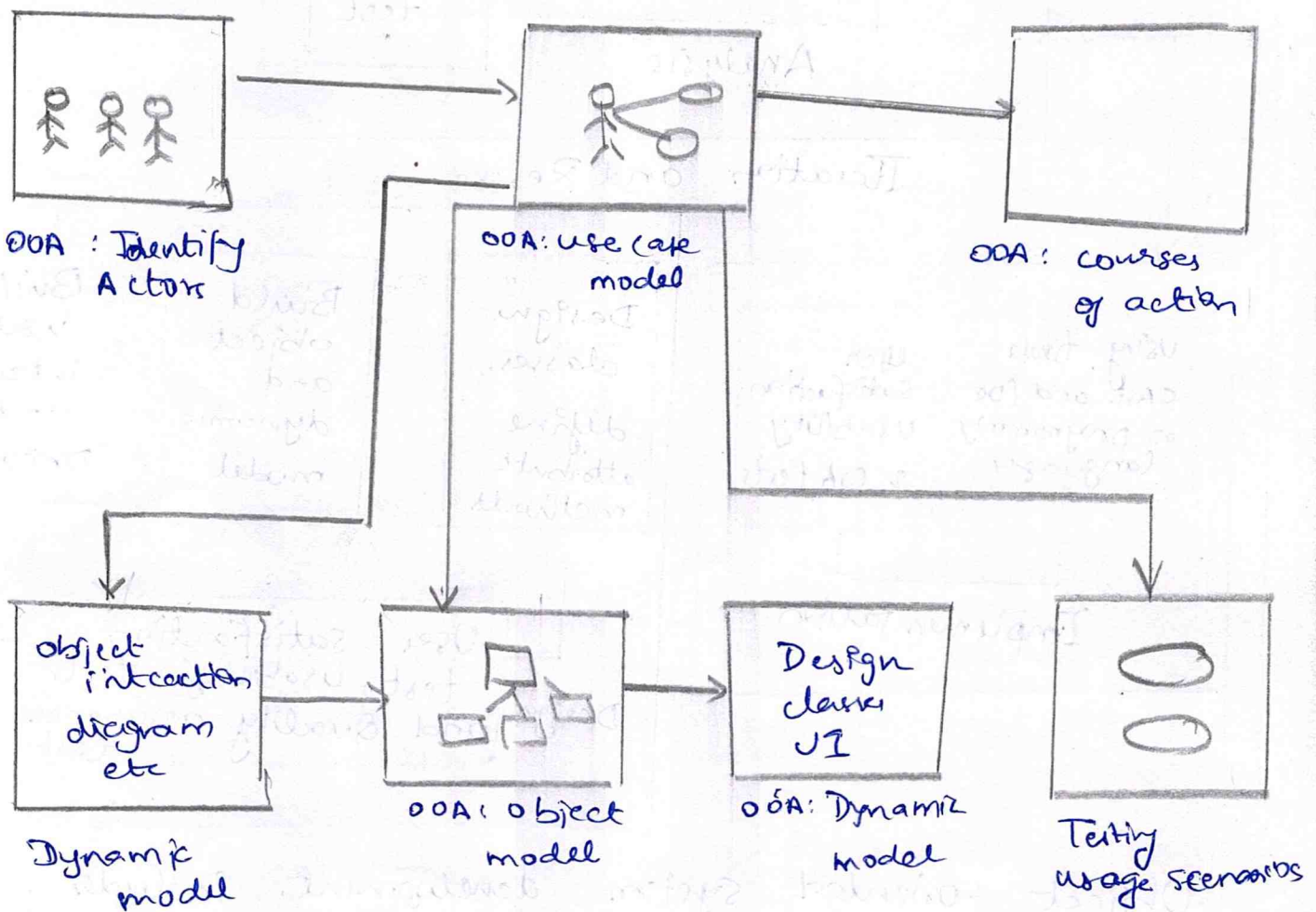
- Object - oriented analysis - use case driven
- Object - oriented design .

- prototyping
- components - based development
- Incremental testing.

Object-oriented Analysis use case driven.

- The object-oriented Analysis phase of software development is concerned with determining the System requirements and identifying classes and their system ~~require~~ relationship to other classes in the problem domain.

To understand the system requirements, we need to identify the user or the actors.



object-oriented design

The goal of object-oriented design (OOD) is to design the classes identified during the analysis phase and the user interface.

- During this phase, we identify and define additional objects and classes that support implementation.

Model

- Design and refine: classes, attributes, methods, structures, association.

Prototyping

- A prototype in object oriented analysis and design is an early working model or sample of a software system or its components.
- It is built to visualize and test certain aspects of system functionality, usability or design aspects before the computer system is fully developed.
- The following categories are some commonly accepted prototypes.
 - Horizontal prototype
 - Vertical prototype
 - Analysis prototypes
 - Domain prototype

Implementation of components - based development

- + Modern manufacturing has evolved to exploit two crucial factors underlying today's market requirements: reduce cost and time to market by building from prebuilt, ready tested components, but add value and differentiation by rapid customization to targeted customer.
- + For example, computer-aided software engineering (CASE) tools allow their user to rapidly develop information systems.

Rapid Application Development (RAD)

Is a set of tools and techniques that can be used to build an application faster than typically possible with traditional methods.

Incremental Testing

- + Incremental testing is a method for evaluating individual modules.
 - After completing unit testing, integration testing ensures that the modules interact with each other correctly.
- + Incremental testing is key in this process, as it integrates modules.